MODULE - II

Beam

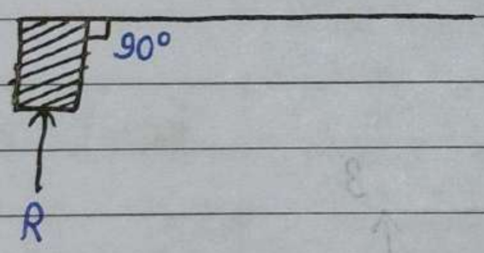
Beam

Beam is a structural member which has one dimension larger than other two dimensions,- Namely depth and width, and it is supported at few points.

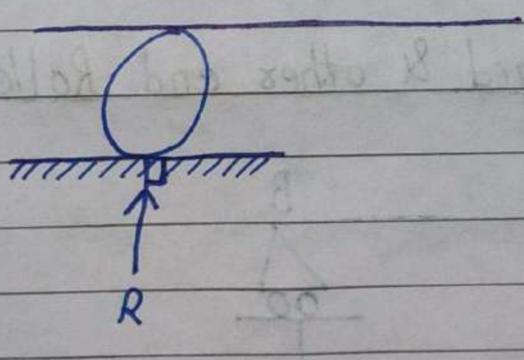
"The distance between two supports is called span."

Types of Support

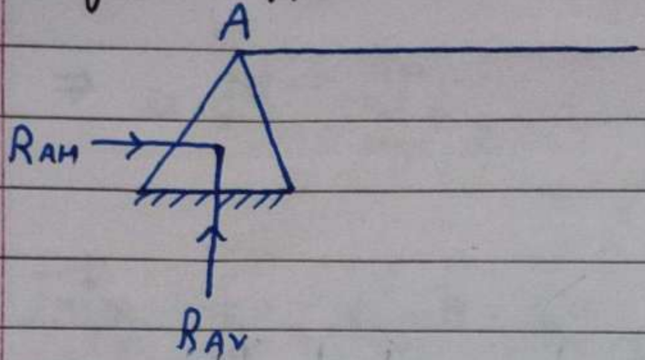
1. Simple Support:



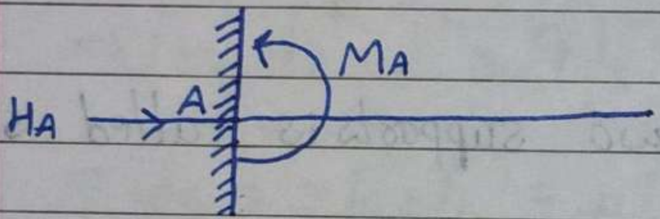
2. Roller Support:



3. <u>Hinged Support:</u>

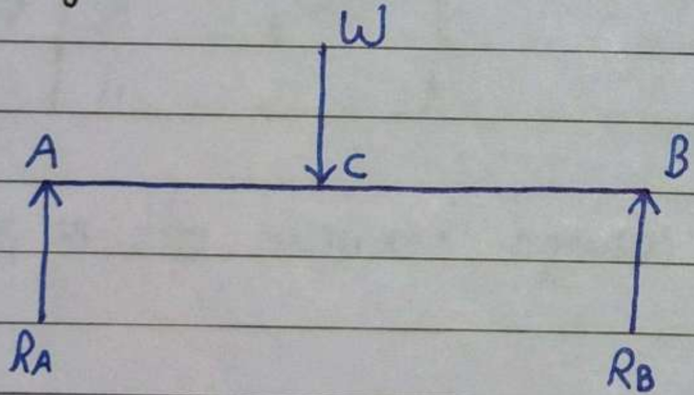


4. <u>Fixed Support:</u>

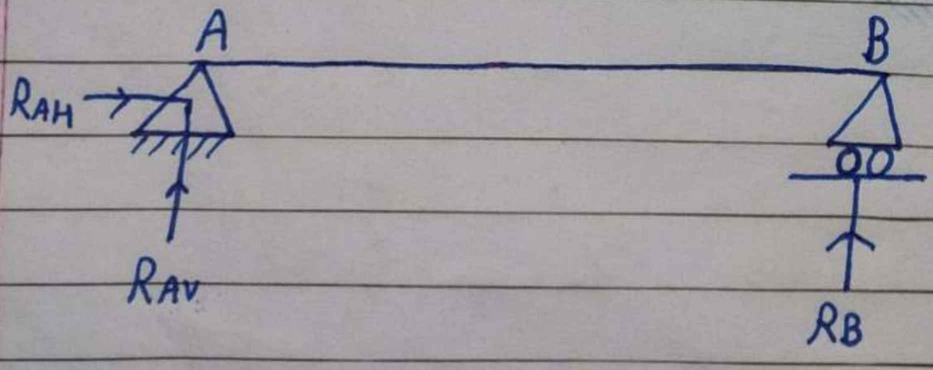


Types of Beam

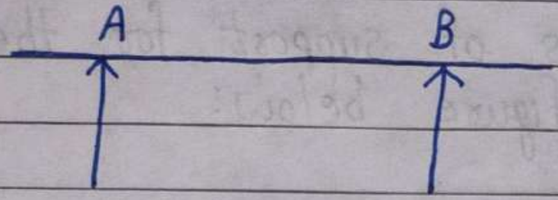
1. <u>Simply Supported Beam:</u>



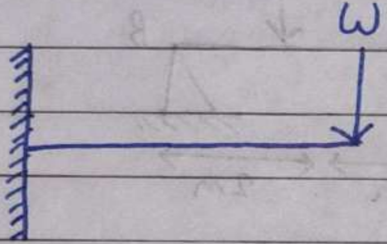
2. <u>Beam with one end Hinged & other end Roller:</u>



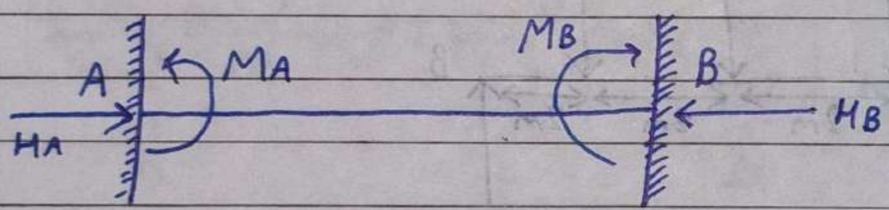
3. Over Hanged Beam:



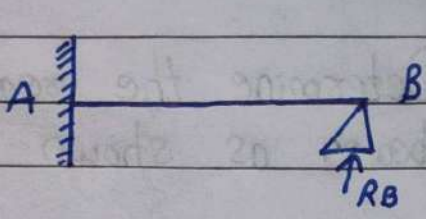
4. Cantilever Beam:



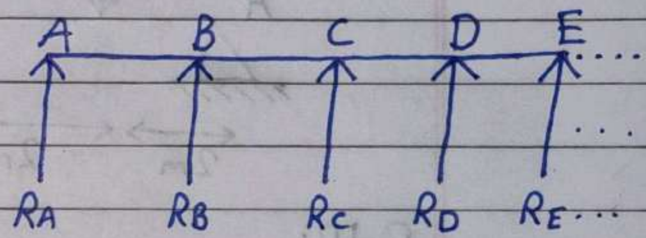
5. Fixed Beam:



6. Propped Cantilever Beam:

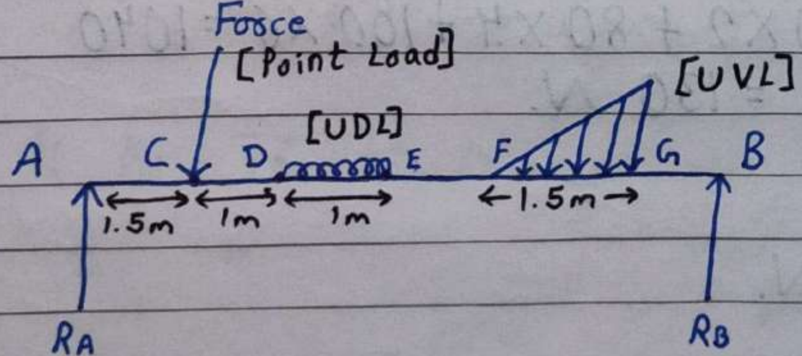


7. Continuous Beam:
[Supports more than two]



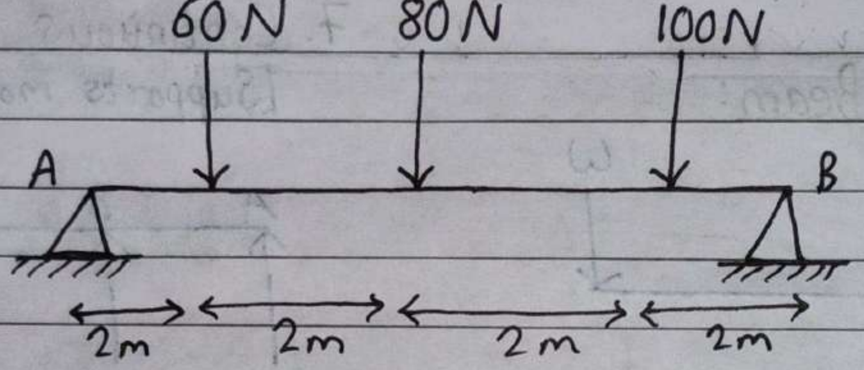
Types of Load

1. Point Load [C]
2. UDL (Uniformly Distributed Load) [D-E]
3. UVL (Uniformly Varying Load) [F-G]

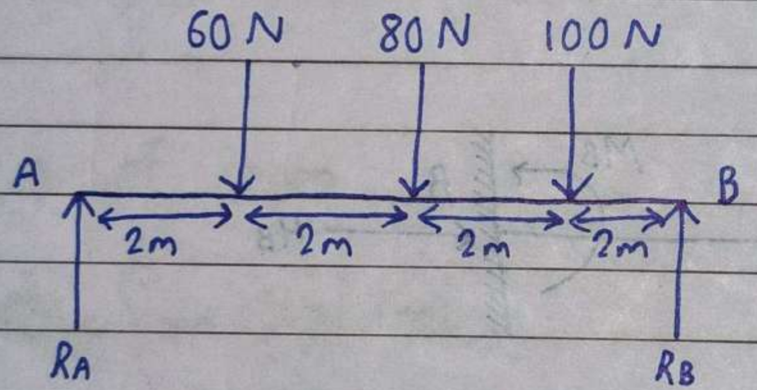


Numericals

Q) Determine the reactions of support for the following beam as shown in figure below:



Sol^n:



$\Sigma F_x = 0$

$\Sigma F_y = 0 \Rightarrow R_A + R_B = 60 + 80 + 100 = 240 \text{ N}$

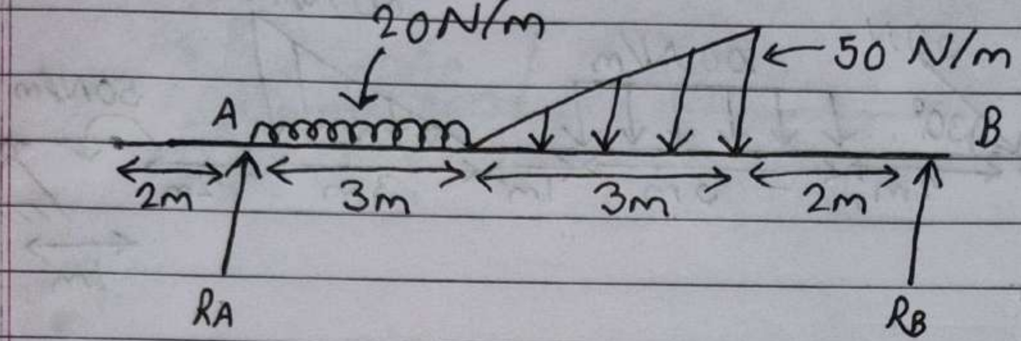
$\Sigma M_A = 0$

$\Rightarrow R_B \times 8 = 60 \times 2 + 80 \times 4 + 100 \times 6 = 1040$

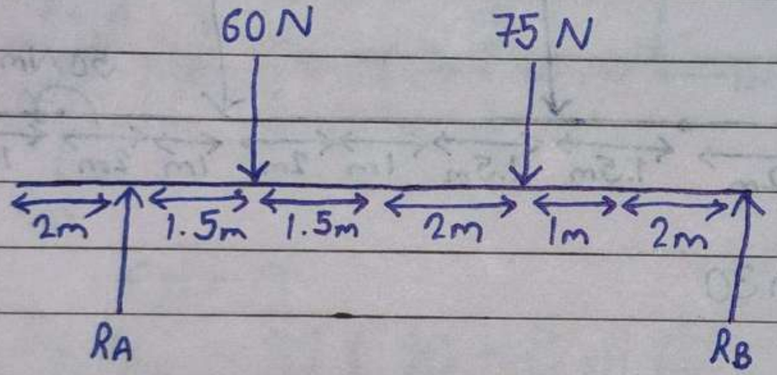
$\Rightarrow R_B = \frac{1040}{8} = 130 \text{ N}.$

$\therefore R_A = 110 \text{ N}.$

Q) Find the reactions at A & B for the following beam as shown in figure below:



Sol^n:



$\Sigma F_y = 0$

$\Rightarrow R_A + R_B = 60 + 75 = 135 \text{ N}$

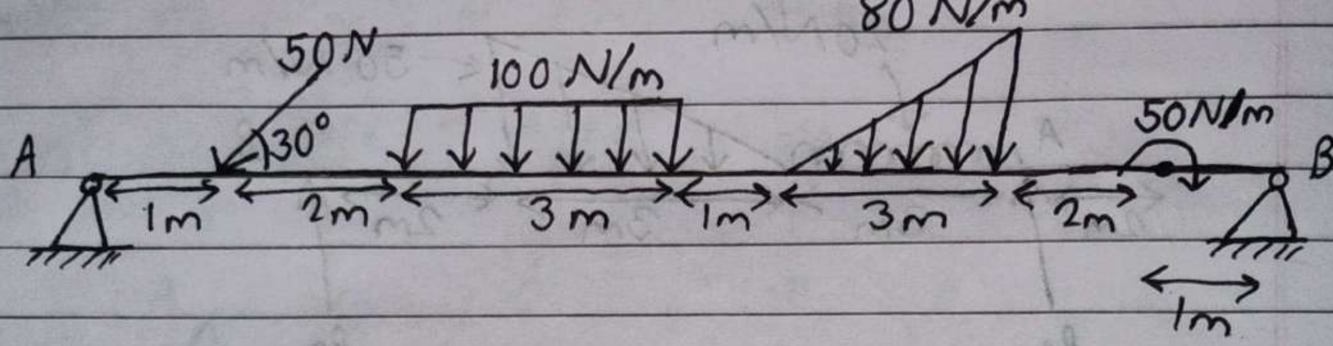
$\Sigma M_A = 0$

$\Rightarrow R_B \times 8 = 60 \times 1.5 + 75 \times 5 = 465$

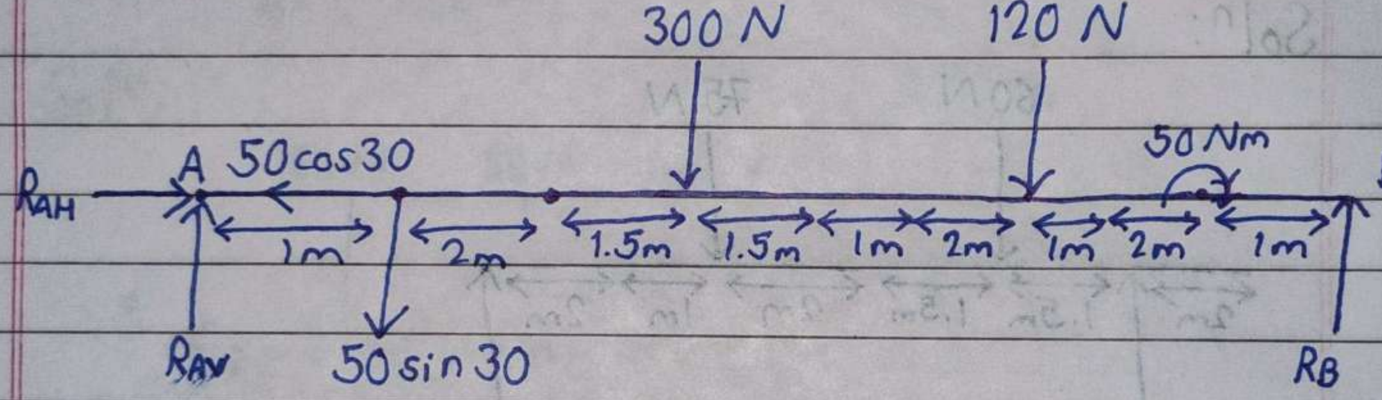
$\therefore R_B = 58.125 \text{ N}$

& $R_A = 76.875 \text{ N}$

Q) Find the reactions at A & B for the following beam as shown in figure below.



Sol^n:



$\Sigma F_x = 0$

$\Rightarrow R_{AH} = 50 \cos 30 = 43.30$ N.

$\Sigma M_A = 0$

$\Rightarrow R_B \times 13 = 1 \times 50 \sin 30 + 300 \times 4.5 + 120 \times 9 + 50$

$\Rightarrow R_B \times 13 = 2505$

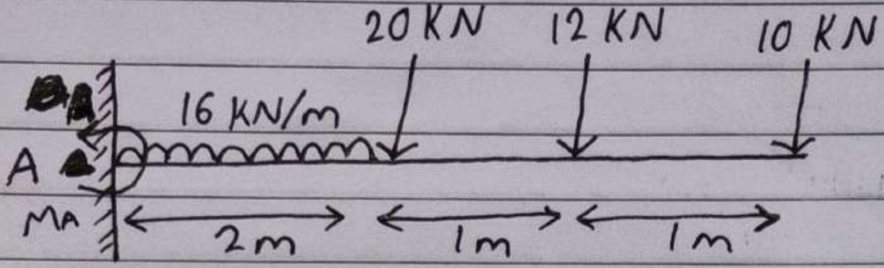
$\therefore R_B = \dfrac{2505}{13} = 192.692$ N.

$\Sigma F_y = 0 \Rightarrow R_{AV} + R_B = 50 \sin 30 + 300 + 120$

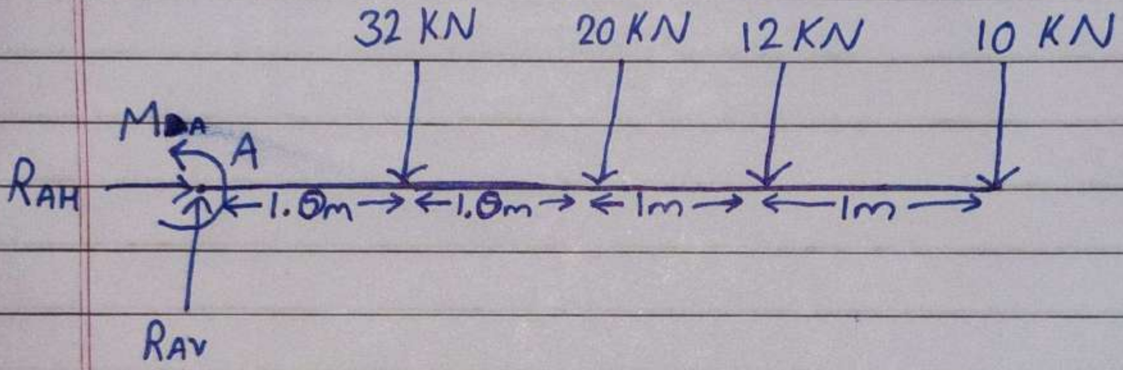
$\Rightarrow R_{AV} + R_B = 445$

$\therefore R_{AV} = 252.308$ N.

Q) Determine the reactions for loaded cantilever beam as shown in figure below.



Sol^n:



$\Sigma Y = 0$

$\Rightarrow R_{AV} = 32 + 20 + 12 + 10 = 74 \text{ KN}.$

& $R_{AH} = 0 \text{ N}.$

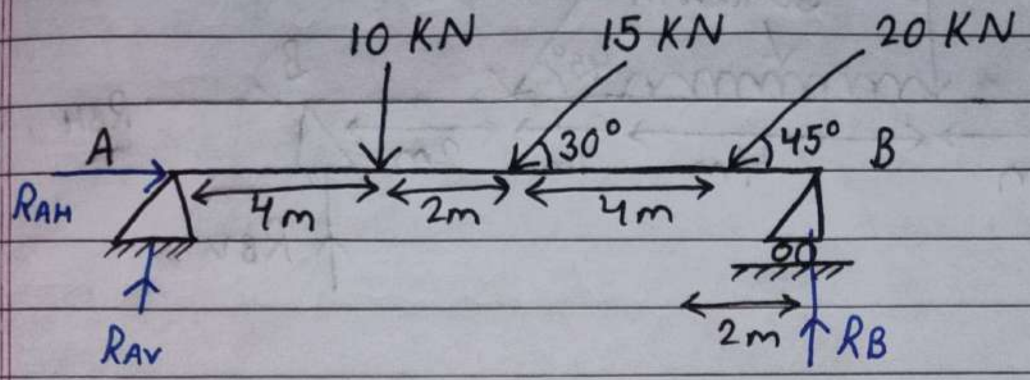
Also, $\Sigma M_A = 0$

$\Rightarrow M_A = 32 \times 1 + 20 \times 2 + 12 \times 3 + 10 \times 4$

$\therefore M_A = 148 \text{ KNm}.$

Numericals from Beam

9)



Sol^n:

$R_{AH} = 15\cos 30 + 20\cos 45 = 27.132\ KN$

$R_{AV} + R_B = 10 + 15\sin 30 + 20\sin 45 = 31.642\ KN$

$\Sigma M_A = 0$

$\Rightarrow 10 \times 4 + 15 \times 6 \sin 30 + 20 \times 10 \sin 45 = R_B \times 12$

$\cancel{\Rightarrow 185 = R_B \times 12}$ | $\Rightarrow 226.421 = R_B \times 12$

$\cancel{\Rightarrow R_B = 15.416\ KN}$ | $\Rightarrow R_B = 18.868\ KN$

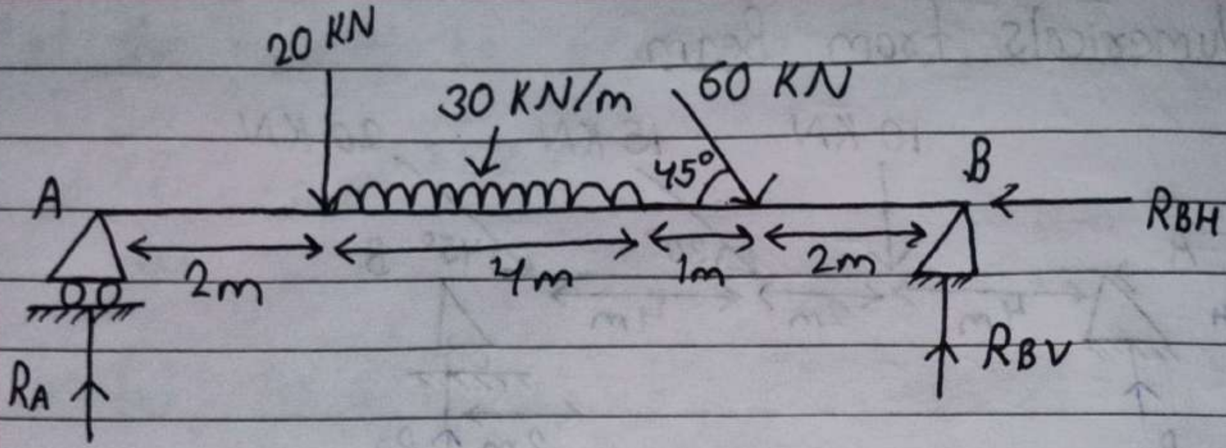
$R_{AV} = 31.642 - 18.868 = \cancel{16.226\ KN} = 12.774\ kN$

$\therefore R_A = \sqrt{R_{AH}^2 + R_{AV}^2} = 29.988\ kN$

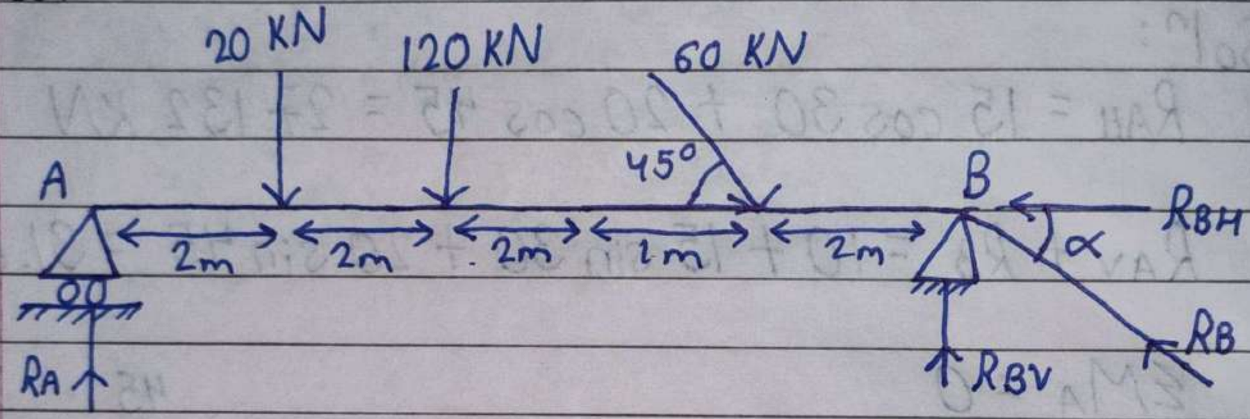
$\tan\theta = \dfrac{R_{AV}}{R_{AH}}$

$\Rightarrow \theta = \tan^{-1}\left(\dfrac{12.774}{27.132}\right) = 25.211°$

9)



Sol^n:



$R_{BH} = +60 \cos 45 = +42.426$ KN

$R_A + R_{BV} = 20 + 120 + 60 \sin 45 = 182.426$ KN

$\Sigma M_A = 0$

$\Rightarrow 20 \times 2 + 120 \times 4 + 60 \times 7 \sin 45 = R_{BV} \times 9$

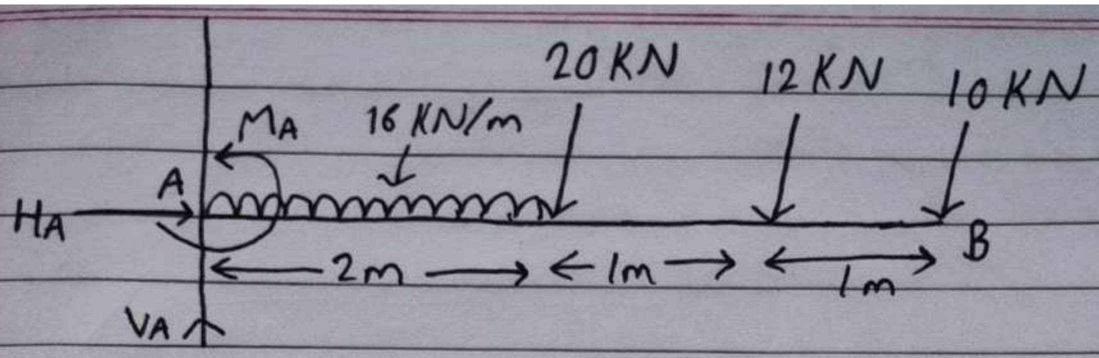
$\Rightarrow R_{BV} = 90.776$ KN

$\therefore R_A = 91.649$ KN

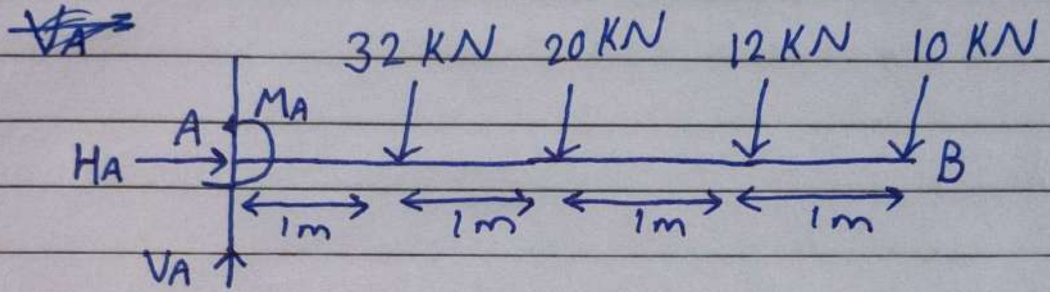
$R_B = \sqrt{R_{BV}^2 + R_{BH}^2} = 100.201$ KN

$\alpha = \tan^{-1}\left(\dfrac{R_{BV}}{R_{BH}}\right) = 64.950°$.

Q)



Sol^n:

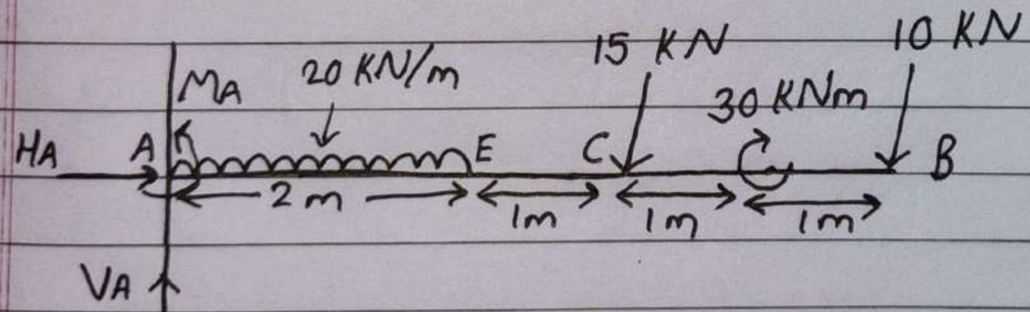


$H_A = 0$

$V_A = 32 + 20 + 12 + 10 = 74\ KN$

$M_A = 32 \times 1 + 20 \times 2 + 12 \times 3 + 10 \times 4 = 148\ KN$

Q)



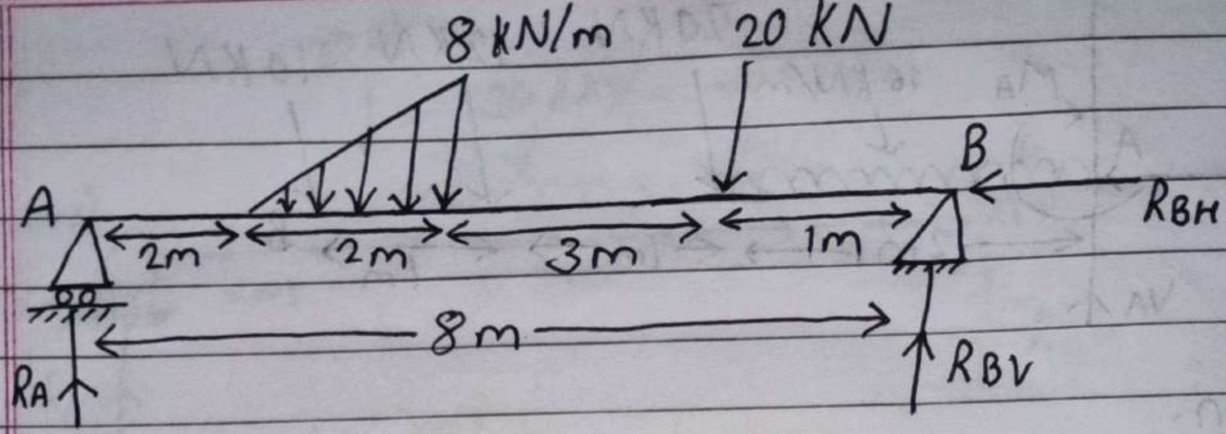
Sol^n:

$H_A = 0$

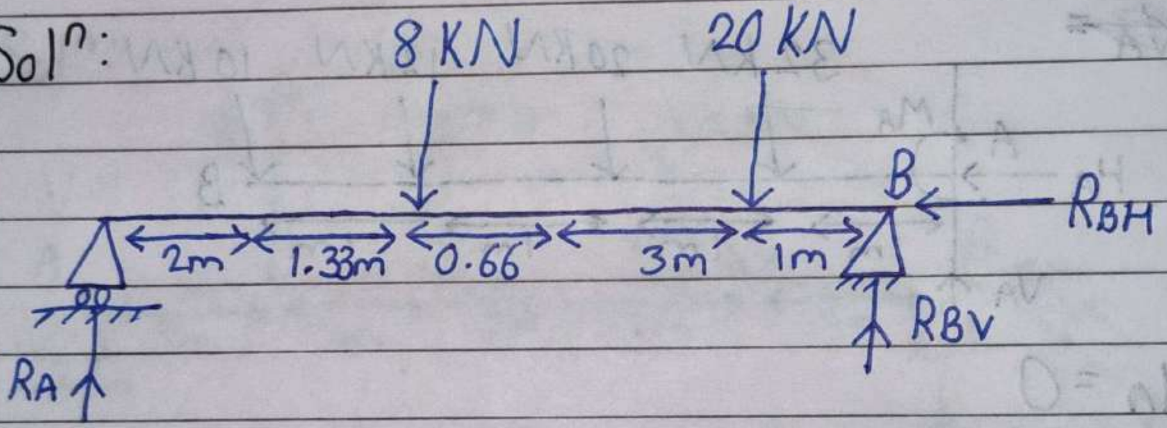
$V_A = 40 + 15 + 10 = 65\ KN$

$M_A = 40 \times 1 + 15 \times 3 + 30 + 10 \times 5 = 165\ KNm.$

9)



Sol^n:



$R_{BH} = 0$

$R_A + R_{BV} = 8 + 20 = 28$

$\Sigma M_A = 0$

$\Rightarrow 8 \times 3.33 + 20 \times 6.99 = R_{BV} \times 8$

$\Rightarrow R_{BV} = 20.805$ KN

$\therefore$ ~~$R_A = 8.805$ KN~~ $\therefore R_A = 7.195$ KN

& $R_B = 20.805$ KN.